

Adaptive Graph Fusion for Conditional Local Convolution: A Schedule-Matched, Multi-Seed Evaluation on WeatherBench

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ABSTRACT

Conditional Local Convolution Recurrent Network (CLCRN) is an established graph-recurrent model for meteorological forecasting. Rather than pursuing a leaderboard result, we ask under controlled conditions whether augmenting this backbone with non-local adaptive structure yields reproducible, architecture-attributable gains, and for which variables. We examine whether a compact adaptive graph-fusion (AGF) encoder can improve CLCRN by learning non-local node dependencies before the original recurrent backbone. The extension builds a directed adjacency from trainable source and destination node embeddings, keeps a renormalized top- k neighbourhood, and fuses the graph-aggregated representation with a residual projection through a learned gate. Experiments use the WeatherBench preprocessing and four tasks from the original CLCRN study: 2-m temperature, relative humidity, 10-m surface wind, and total cloud cover. We compare CLCRN-AGF with a schedule-matched CLCRN control over five seed replicates. The clearest effect is relative humidity: CLCRN-AGF reduces mean absolute error by 2.28% and root mean squared error by 3.02%; all seeds improve (Welch tests: $p = 7.6 \times 10^{-4}$ and $p = 8.4 \times 10^{-5}$). Effects on temperature, wind, and cloud cover are small or metric-dependent. Component ablations show that the adaptive graph branch contributes most clearly for humidity, whereas replacing the learned gate by mean fusion remains competitive. These results narrow the contribution of adaptive graph fusion: it is useful for the humidity task but is not a uniformly beneficial replacement for the original CLCRN encoder. More broadly, the study shows how a schedule-matched control, multi-seed replication, and significance testing separate genuine architectural gains from optimization and reporting effects—a discipline directly transferable to other spatio-temporal forecasting models.

1 Introduction

Numerical weather prediction (NWP) remains the operational standard for atmospheric forecasting because it explicitly integrates the governing physical equations¹. Its accuracy, however, comes at a substantial computational cost, and recent years have seen data-driven models emerge as fast and increasingly capable alternatives for selected forecasting tasks. The release of WeatherBench established a common benchmark built on the ERA5 reanalysis that made systematic comparison of such models possible^{2,51}, and early convolutional sequence models on regular or cubed-sphere grids demonstrated that deep networks can learn skillful short-range dynamics³. At the large-model end of the spectrum, FourCastNet, Pangu-Weather, GraphCast, ClimaX, spherical Fourier neural operators, neural general circulation models, and probabilistic diffusion forecasters now rival or exceed operational NWP at global scale^{4–10}. Smaller graph-recurrent systems nonetheless remain relevant whenever retraining cost, task-specific adaptation, controlled ablation, and reproducibility matter more than raw global skill.

Because the atmosphere is sampled on an irregular spherical grid rather than a Euclidean image lattice (Fig. 1), graph neural networks (GNNs) are a natural inductive bias for these data. On the curved manifold, geodesic distance, relative orientation, and spatial scale all vary across the globe, so a single shared convolution kernel is geometrically mismatched and geometry-unaware models are prone to overfitting; geometry-conditioned graph convolution with appropriate regularization is therefore better suited to the data. Directly relevant to the present study, the Conditional Local Convolution Recurrent Network (CLCRN)¹¹ introduced a Conditional Local Convolution (CLConv) that generates *location-dependent* kernels from spherical coordinates, geodesic distance, and directional information, thereby addressing the spatially shared-kernel limitation of conventional graph convolution on an unstructured spherical grid. We review the broader spatio-temporal graph, attention-based, and weather-forecasting literature in Sections 1.1–1.3.

Despite this progress, an open question persists for geometry-aware backbones such as CLCRN. CLConv already encodes geographically local structure through its coordinate-conditioned kernels, yet many meteorological variables exhibit dependencies that are not fully described by spatial proximity—moisture transport, large-scale circulation, and teleconnections couple regions that are geodesically distant. It is therefore unclear whether adding a learned, *non-local* adaptive graph on

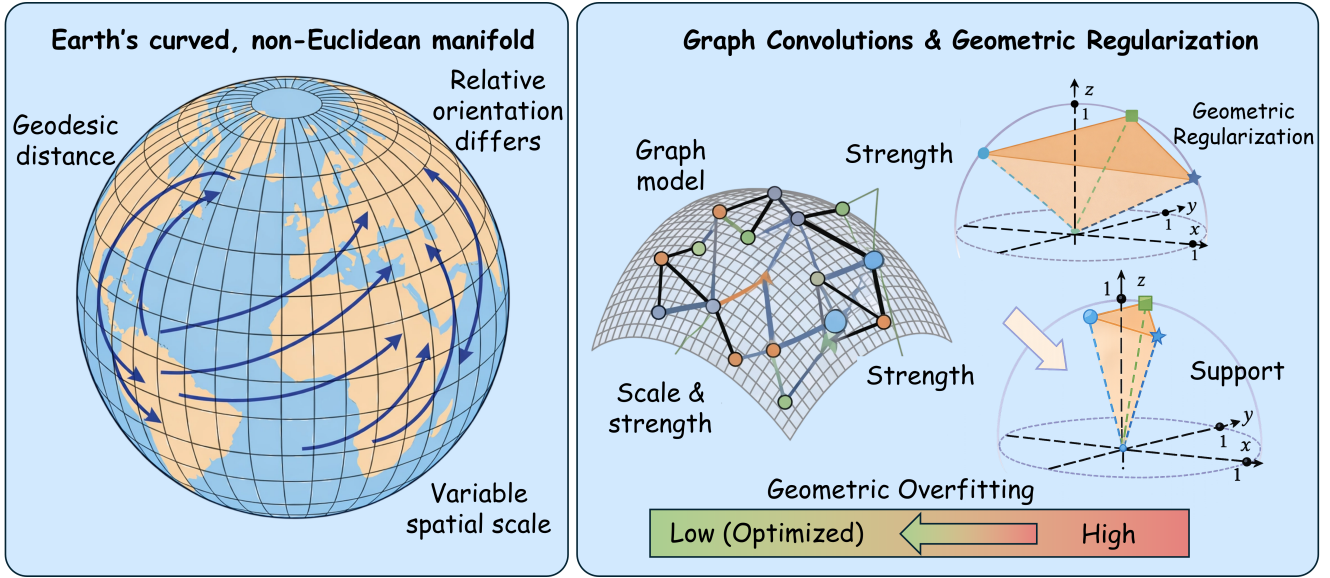


Figure 1. Motivation: forecasting on the Earth’s curved, non-Euclidean manifold. Left: on the spherical surface, geodesic distance, relative orientation, and spatial scale all vary with location, so a single convolution kernel shared across the grid is geometrically mismatched. Right: a graph model built on the curved mesh must adapt the scale and strength of its connections to each neighbourhood. Geometric regularization narrows a kernel supported on a wide region of the unit sphere (top) to a compact local support (bottom), driving geometric overfitting from high to low. This motivates the geometry-conditioned CLConv backbone retained in this study, and raises the question of what a learned non-local graph adds on top of it.

top of a backbone that is already geometry-conditioned yields genuine and reproducible gains, or merely adds parameters. Two confounds make this hard to answer in practice. First, reported improvements often entangle architectural changes with optimization and software-environment differences, so single best-run comparisons can be misleading. Second, gains can be task-specific, and aggregate averages hide which variables actually benefit. Resolving these issues requires a schedule-matched control, multiple random-seed replicates with explicit significance testing, and a clean separation of published reference values from newly generated results—an evaluation discipline that the recent benchmarking literature increasingly emphasizes.

To address this question we propose **CLCRN-AGF** (Fig. 2), which inserts a compact adaptive graph-fusion encoder before the original CLCRN recurrent backbone, leaving the backbone unchanged. Trainable source and destination node embeddings define a directed similarity matrix; a sparse top- k neighbourhood aggregates node features; and a learned gate mixes the graph-aggregated representation with a residual projection. The design tests the focused hypothesis that non-local learned dependencies add value beyond CLCRN’s geographically conditioned local convolution, particularly for variables whose dynamics are not fully described by local proximity. The study is deliberately framed as a controlled evaluation and reproducibility analysis rather than a leaderboard entry, and its primary contribution is methodological.

The main contributions of this paper are as follows:

- **Controlled Evaluation Protocol.** A schedule-matched control, five-seed replication, confidence intervals, Welch tests, false-discovery-rate correction, and a distribution-free check that together separate architectural effects from optimization and reporting effects.
- **Adaptive Graph-Fusion (AGF) Encoder.** The compact adaptive top- k graph-fusion encoder that is the object of this evaluation, attached to the established CLCRN backbone while leaving the backbone unchanged.
- **Component Ablations and Diagnostics.** Ablations separating the adaptive graph branch from the learned fusion gate, together with gate-activation, robustness, and convergence diagnostics.
- **Transparent Reporting.** A transparent separation between published reference values, independent reproduction, and newly generated results.
- **A Positive as well as Negative Result.** The extension consistently helps relative humidity but gives small or metric-dependent changes for the other tasks.

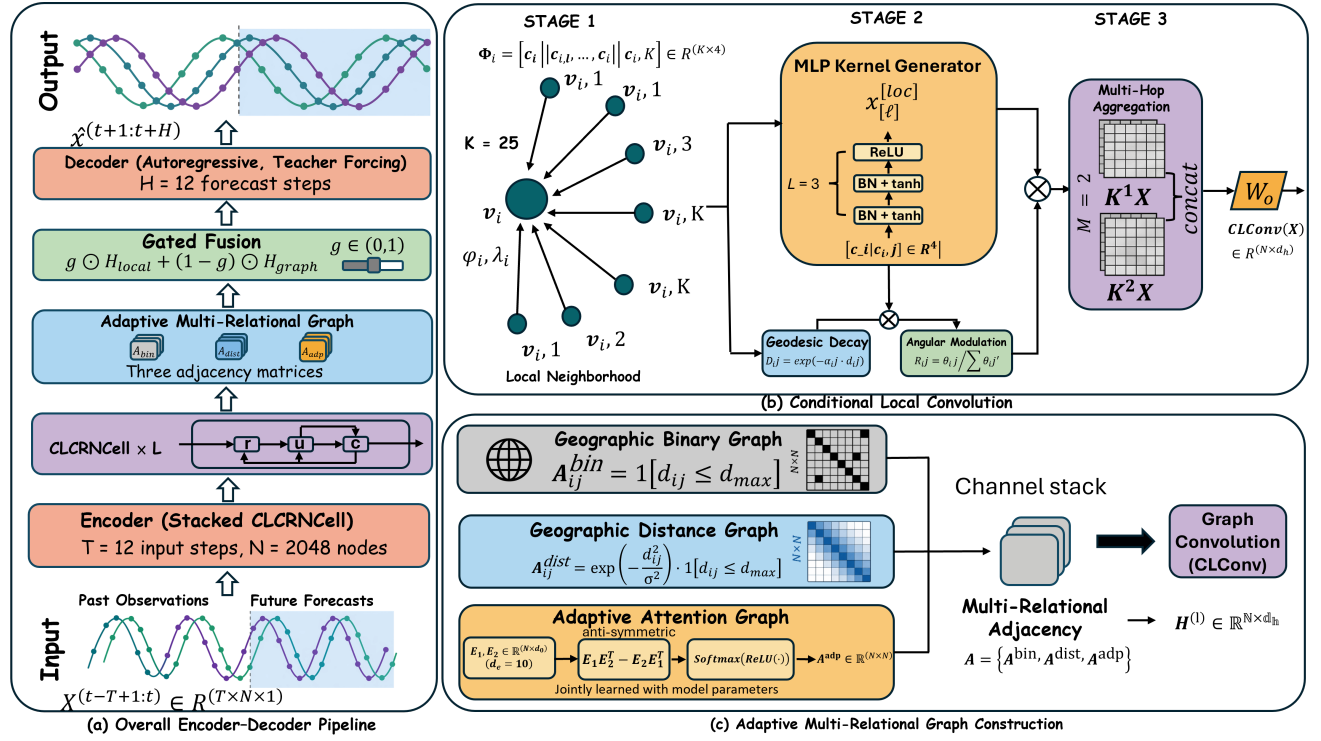


Figure 2. Architecture of the evaluated CLCRN-AGF model. (a) Overall encoder–decoder pipeline. A 12-step input sequence on the $N = 2,048$ -node spherical grid is processed by a stacked-CLCRNCell encoder; a learned gate $g \in (0, 1)$ fuses the geometry-conditioned local representation with the adaptive graph representation, and an autoregressive decoder with teacher forcing emits the 12-step forecast. (b) Conditional Local Convolution (CLConv), the unchanged backbone operator of Lin et al.¹¹. For each node, an MLP kernel generator conditioned on centre–neighbour spherical coordinates produces a location-dependent kernel over the $K = 25$ -node local neighbourhood; the kernel is modulated by geodesic decay and angular normalization and applied through multi-hop aggregation followed by an output projection. (c) Construction of the relational structure. Geographic binary and distance-weighted neighbourhoods encode the local spherical topology exploited by the CLConv backbone, whereas the adaptive attention graph—learned jointly with the model parameters from trainable source and destination node embeddings ($d_e = 10$) and sparsified by top- k selection (Methods)—supplies the learned non-local dependencies whose contribution this study isolates.

1.1 Spatio-temporal graph neural networks: from fixed to learned topology

Three strands of prior work frame the design summarized in Fig. 2. The first supplies the graph-convolutional machinery itself: spectral and spatial graph convolutions^{12,13}, graph attention¹⁴, and inductive neighborhood aggregation¹⁵ established the basic operator toolbox, while gauge-equivariant convolutions designed for icosahedral and spherical meshes⁵² brought that toolbox to the curved domains that weather data inhabit.

Building on these operators, spatio-temporal GNNs (STGNNs) for structured forecasting initially consumed a fixed, pre-computed topology—diffusion convolution within recurrent encoders¹⁶, fully convolutional spatio-temporal graphs¹⁷, and attention-augmented graph convolution¹⁸ all operate on distance-thresholded geographic graphs of the kind shown in Fig. 2c (binary and distance-weighted adjacency). A second wave learned the topology itself: self-adaptive adjacency¹⁹, node-adaptive parameter learning²⁰, jointly learned mix-hop graphs²¹, multi-attention networks²², continuous-time graph ODEs²³, and dynamic spatial–temporal aware graphs²⁴ infer edges from trainable node embeddings, the lineage that the adaptive attention graph in Fig. 2c follows. Comprehensive surveys document this trajectory²⁵, and recent work extends graph forecasting to scalable, probabilistic, state-space, and irregularly-sampled settings^{57–60}. Most of these STGNNs, however, assume a Euclidean or generic graph domain; geometry-conditioned operators such as CLConv (Fig. 2b) remain comparatively rare, which motivates retaining CLCRN as the fixed, unmodified backbone of this study.

1.2 Attention-based and long-sequence forecasting models

A second strand removes the pre-specified graph altogether and lets attention or probabilistic structure learning discover the dependency pattern. Examples include discrete graph-structure inference²⁶, position- and edge-conditioned processing on point

clouds^{55,56}, and propagation-delay-aware or adaptive-embedding transformers for spatio-temporal data^{27–30}. General-purpose long-sequence forecasters point the same way: the Transformer³¹ and its forecasting descendants Informer³², Autoformer³³, PatchTST³⁵, TimesNet³⁶, Crossformer³⁷, and iTransformer³⁸, alongside selective state-space models³⁹, spectral graph forecasters⁴⁰, pure deep stacks such as N-BEATS⁴¹, and the deliberately simple DLinear³⁴, repeatedly show that learned, non-local dependency structure can substitute for a pre-specified topology. This is precisely the hypothesis that the gated fusion of local and graph representations in Fig. 2a is designed to test on a geometry-conditioned backbone: if non-local learned structure carries signal that the local CLConv kernel misses, the gate has room to exploit it.

1.3 Deep learning for weather and climate prediction

The third strand is the application domain itself. Within *Scientific Reports*, adaptive and multi-scale spatio-temporal graph networks have become a recurring theme for traffic and environmental forecasting^{47–49}, as have deep models for climate and microclimate prediction^{42–44,46}. The present study follows this trajectory but inverts the usual emphasis: rather than proposing a larger model, it treats the geometry-conditioned CLCRN backbone of Fig. 2b as a fixed substrate and asks, under a controlled, schedule-matched, multi-seed protocol, whether the learned non-local graph of Fig. 2c adds reproducible, architecture-attributable value.

2 Results

2.1 Experimental setup

2.1.1 Datasets

We evaluate on the four WeatherBench-derived tasks defined by the original CLCRN study^{2,11}: 2-m temperature (K), relative humidity (percentage points), 10-m surface wind components (u, v ; m s^{-1}), and total cloud cover (fraction). Hourly ERA5 records from 1 January 2010 to 31 December 2018 are mapped to a 32×64 latitude–longitude grid with $N = 2,048$ nodes. Each example uses 12 consecutive hourly observations to forecast the next 12 hours; starting windows are sampled once per day, yielding 3,286 examples split chronologically into 70%/10%/20% for train/validation/test.

2.1.2 Baselines and comparison protocol

The primary, causal comparison is between CLCRN-AGF and a *schedule-matched control*: the original CLCRN backbone trained with the identical data, optimizer, training budget, and random seeds but with the AGF module disabled. As external context we also report (i) persistence on the current split; (ii) the published CLCRN reference values¹¹; (iii) a single independent reproduction of the public CLCRN implementation; and (iv) the family of graph forecasting baselines benchmarked in the original CLCRN study—STGCN¹⁷, DCRNN¹⁶, AGCRN²⁰, ASTGCN and MSTGCN¹⁸, a graph-convolutional GRU⁵³, and a temporal GCN⁵⁴—whose relative standing against the CLCRN backbone is summarized in Fig. 3. These external numbers combine architecture, software, and protocol effects and are therefore used only to situate the backbone, not to attribute the AGF effect.

2.1.3 Evaluation metrics and significance

We report mean absolute error (MAE) and root mean squared error (RMSE) computed after inverse standardization, using a corrected zero-inclusive evaluator (zero wind and zero cloud cover are physically meaningful and are not treated as missing). The primary analysis uses five random-seed replicates $\{2021, \dots, 2025\}$; for each metric we report the mean and standard deviation across seeds, the mean difference, a two-sided 95% Welch confidence interval, and a two-sided Welch test.

2.1.4 Implementation details

Both variants use two recurrent layers with 32 hidden units; CLCRN-AGF adds $d_e = 10$ -dimensional adaptive source/destination node embeddings, an 8-dimensional AGF output, and $k = 4$ selected neighbours (self-edge included). Training uses Adam (learning rate 0.01, batch size 32, gradient clipping 5) with a MultiStepLR schedule for up to 100 epochs, selecting the lowest-validation-MAE checkpoint. All runs use Python 3.13, PyTorch 2.10, CUDA 12.8, and a single NVIDIA GeForce RTX 5090 Laptop GPU. Full hyperparameters are listed in Methods.

2.2 Backbone standing among graph baselines

Before evaluating the extension, Fig. 3 situates the CLCRN backbone relative to seven graph forecasting baselines reported in the original CLCRN benchmark, across all four variables and both metrics. The backbone is consistently strong, which is the reason it is adopted here as a fixed substrate: the present study does not claim CLCRN or CLConv as new contributions, and these external values are shown only as context for the backbone rather than as the causal test of AGF.

Main Results on WeatherBench against Seven Graph Forecasting Baselines

	Temperature MAE	Temperature RMSE	Humidity MAE	Humidity RMSE	Wind MAE	Wind RMSE	Cloud Cover MAE	Cloud Cover RMSE	
DCRNN	1.872	2.951	8.317	12.840	2.139	3.276	0.228	0.358	Best Middle Worst
AGCRN	1.612 #2	2.576 #2	6.843 #2	10.870 #2	1.824 #2	2.893 #2	0.192 #2	0.311 #2	
STGCN	2.234	3.412	9.404	14.820	2.577	3.891	0.264	0.402	
GC-GRU	1.948	3.124	8.231	12.850	2.227	3.410	0.235	0.367	
ASTGCN	1.818	2.897	7.652	12.060	2.098	3.248	0.222	0.354	
MSTGCN	1.893	3.049	7.909	12.480	2.173	3.351	0.230	0.362	
Temporal GCN	2.147	3.283	8.571	13.560	2.408	3.670	0.247	0.385	
CLCRN (Ours)	1.169 #1	1.883 #1	4.531 #1	7.078 #1	1.326 #1	2.129 #1	0.149 #1	0.246 #1	
Cell color is rank-normalised within each dataset/metric column: greener = better, redder = worse.									

Figure 3. Standing of the CLCRN backbone among seven graph forecasting baselines on the four WeatherBench tasks. Cells are rank-normalized within each metric column (greener is better). These values are reported by the original CLCRN benchmark and are shown only to situate the backbone; they are not the causal comparison for the AGF extension.

2.3 Primary replicate comparison

The primary comparison uses five random-seed replicates. The control is the original CLCRN backbone trained with the same revised schedule as CLCRN-AGF. Table 1 reports mean and standard deviation across seeds, the mean difference (CLCRN-AGF minus control), a 95% Welch confidence interval, and a two-sided Welch test. The same seed labels are used in both queues, but the AGF module changes model initialization order; the analysis is therefore treated as a replicate comparison rather than a strict common-random-number paired design.

Table 1. Five-replicate comparison of CLCRN-AGF and the schedule-matched CLCRN control. Negative differences favour CLCRN-AGF. Confidence intervals and p -values use Welch’s unequal-variance approximation.

Task	Metric	Control ($\mu \pm \sigma$)	CLCRN-AGF ($\mu \pm \sigma$)	Difference [95% CI]	p_{Welch}
Temperature	MAE	1.229 ± 0.057	1.233 ± 0.032	$+0.004 [-0.067, +0.075]$	0.8985
	RMSE	2.097 ± 0.073	2.020 ± 0.047	$-0.076 [-0.169, +0.016]$	0.0921
Relative humidity	MAE	4.633 ± 0.032	4.528 ± 0.013	$-0.106 [-0.144, -0.067]$	7.6×10^{-4}
	RMSE	7.237 ± 0.051	7.019 ± 0.040	$-0.218 [-0.285, -0.151]$	8.4×10^{-5}
Surface wind	MAE	1.244 ± 0.005	1.240 ± 0.007	$-0.004 [-0.014, +0.006]$	0.3495
	RMSE	1.977 ± 0.008	1.973 ± 0.017	$-0.004 [-0.024, +0.017]$	0.6851
Cloud cover	MAE	0.1487 ± 0.0007	0.1480 ± 0.0013	$-0.0007 [-0.0023, +0.0009]$	0.3106
	RMSE	0.2473 ± 0.0019	0.2470 ± 0.0027	$-0.0003 [-0.0038, +0.0031]$	0.8368

Relative humidity is the only task for which every seed label favours CLCRN-AGF for both metrics (Fig. 4). Its mean MAE decreases by 2.28% and RMSE by 3.02%. Temperature shows a metric-dependent result: MAE is essentially unchanged, whereas mean RMSE decreases by 3.64% with an interval that includes zero. Wind and cloud-cover changes are small relative to seed variability.

2.4 Statistical significance analysis

Because eight task/metric comparisons are evaluated jointly, we control the false discovery rate across them with the Benjamini–Hochberg procedure. Only the two relative-humidity improvements remain significant after correction (adjusted $p = 6.7 \times 10^{-4}$ for RMSE and 3.0×10^{-3} for MAE), while every other comparison—including the temperature RMSE difference (adjusted $p = 0.25$)—is not significant (adjusted $p \geq 0.25$). As a distribution-free check that does not rely on the normality assumption at $n = 5$, a one-sided sign test on the humidity replicates—all five seeds favour CLCRN-AGF for both metrics—gives $p = 0.031$. The humidity effect is therefore the only one that survives both multiple-comparison correction and a non-parametric test. These results demonstrate that the observed humidity gain is not caused by random initialization.

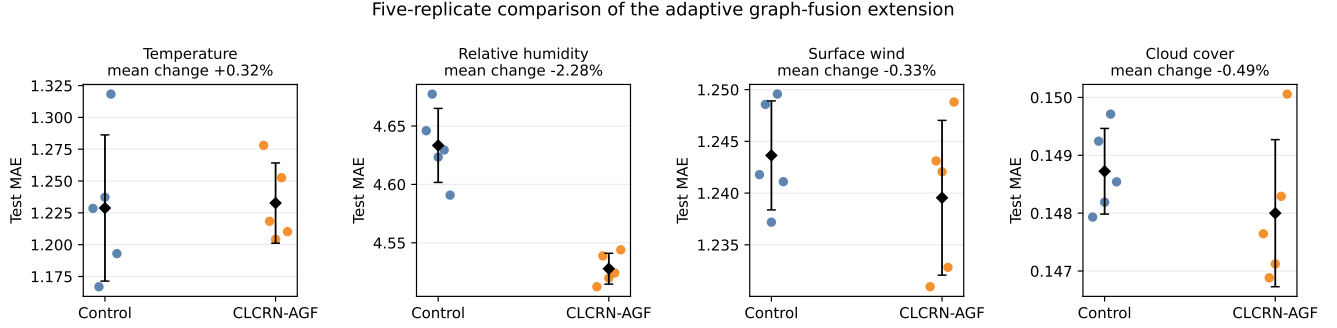


Figure 4. Test MAE for the schedule-matched control and CLCRN-AGF. Dots show individual seed replicates; diamonds show mean $\pm$ one standard deviation. The humidity improvement is consistent across all five seed labels, whereas the other tasks show overlapping or mixed changes.

2.5 Per-step error growth

Figure 5 reports how exact and cumulative error grow with the forecast horizon for each task. As expected for short-range forecasting, error increases monotonically from $t+1$ to $t+12$, but the shape of the growth is variable-specific: temperature and wind accumulate error close to linearly, humidity follows a concave-up profile driven by nonlinear moisture transport, and cloud cover grows sub-linearly beyond roughly $t+4$, consistent with semi-periodic cloud cycles. This profile indicates that the humidity advantage in Table 1 is not an artifact of a single lead time but accrues across the horizon where errors are largest.

2.6 Contextual benchmarks

Table 2 distinguishes results generated in this study from external published values. Persistence is evaluated directly on the current test split. “Published CLCRN” is copied from Lin et al.¹¹ after restoring the humidity and cloud-cover reporting scales used by the current evaluation code. “Reproduced CLCRN” is a single independent run of the public implementation. The two final rows are five-seed results generated under the revised schedule.

Table 2. Contextual test results. Published CLCRN values are external references, not newly generated results. Reproduced CLCRN is a single run; control and CLCRN-AGF are five-seed means. Lower is better.

Method	Temperature		Relative humidity		Surface wind		Cloud cover	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Persistence (current split)	1.421	2.700	6.425	10.447	1.537	2.535	0.1620	0.2725
Published CLCRN ¹¹	1.169	1.883	4.531	7.078	1.326	2.129	0.1491	0.2456
Reproduced CLCRN (one seed)	1.192	2.246	4.677	7.309	1.384	2.266	0.1564	0.2574
Schedule-matched control (five seeds)	1.229	2.097	4.633	7.237	1.244	1.977	0.1487	0.2473
CLCRN-AGF (five seeds)	1.233	2.020	4.528	7.019	1.240	1.973	0.1480	0.2470

The contextual comparison shows why the extension should not be described as uniformly superior. CLCRN-AGF improves on the published reference for relative humidity and wind, is mixed for cloud cover, and does not improve temperature. Differences from the external reference also combine architecture, software environment, and training-protocol effects; the schedule-matched control in Table 1 is therefore the primary causal comparison.

2.7 Component ablations

The experiment archive also contains component ablations. Table 3 separates three kinds of evidence: 100-epoch, three-seed ablations for temperature and relative humidity, and a 20-epoch single-seed screening run over all four tasks. The full-budget humidity comparison uses seeds 2023–2025. The w/o gated fusion humidity completion was trained with automatic mixed precision after repeated standard-precision runs exhausted the local CUDA allocator; checkpoint selection and test evaluation still use the same corrected zero-inclusive evaluator.

In the 100-epoch temperature ablation, removing the entire adaptive graph branch worsens RMSE relative to the full model, while replacing the learned gate by mean fusion gives similar or slightly lower mean errors. Thus the gate is not uniformly beneficial for temperature. In the full-budget humidity ablation, both adaptive-graph variants improve on the model without the adaptive graph branch. The full gated model has the lowest mean MAE and RMSE, but the mean-fusion variant remains close

Per-Step Forecast Error vs. Prediction Horizon

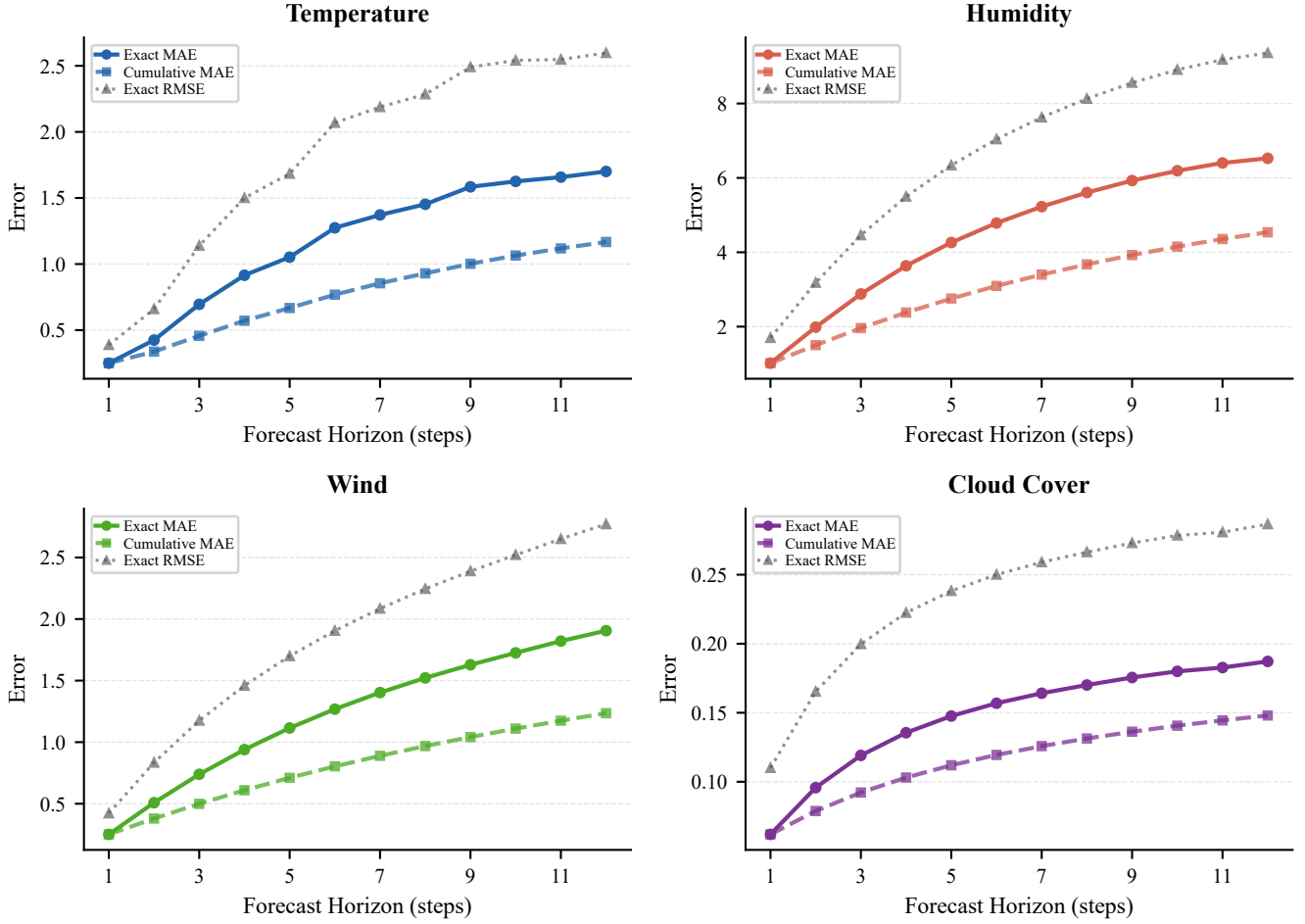


Figure 5. Per-step forecast error against prediction horizon. Exact MAE (solid), cumulative MAE (dashed), and exact RMSE (dotted) for the four tasks over the 12-step (12-hour) horizon. Error grows monotonically with lead time; humidity and temperature exhibit the steepest accumulation.

and was completed under the mixed-precision caveat described above. We therefore treat this block as sensitivity evidence that supports adaptive graph mixing, not as a definitive precision-matched test of the gate. The 20-epoch screening run reaches the same qualitative warning: relative humidity is the most sensitive task, whereas wind, temperature, and cloud-cover screening results are mixed.

2.8 Decomposing the improvement: schedule versus structure

A central reproducibility concern is that apparent architectural gains may in fact originate from the training schedule. Figure 6 decomposes the test MAE of four configurations—an original reproduction, the schedule-matched control (revised schedule, no adaptive branch), the full CLCRN-AGF model, and the published paper target. The revised schedule alone accounts for most of the improvement on wind and cloud cover, with the additional AGF contribution being small; for humidity, the adaptive branch adds a further, consistent reduction on top of the schedule. This decomposition is the reason the five-replicate, schedule-matched control—rather than a single best run—is treated as the primary causal comparison throughout.

2.9 Reproducibility, stability, and convergence

Because the schedule-matched comparison is the basis of our conclusions, we characterize its run-to-run stability directly. Across the five seeds in Table 1, CLCRN-AGF is highly repeatable: the per-seed standard deviation of test MAE is at most 0.032 (temperature) and is much smaller for the other variables (0.013 for humidity, 0.007 for wind, and 0.0013 for cloud

Table 3. Component ablations. The first two blocks report 100-epoch three-seed ablations at the full 12-hour horizon; the final block reports a 20-epoch single-seed screening ablation for all four tasks. Lower is better.

Budget	Task	Variant	MAE	RMSE
100 epochs, 3 seeds	Temperature	w/o adaptive graph	1.265 ± 0.046	2.143 ± 0.024
		w/o gated fusion	1.189 ± 0.098	2.018 ± 0.079
		Full CLCRN-AGF	1.222 ± 0.026	2.032 ± 0.062
100 epochs, 3 seeds	Relative humidity	w/o adaptive graph	4.615 ± 0.021	7.223 ± 0.043
		w/o gated fusion	4.575 ± 0.042	7.077 ± 0.076
		Full CLCRN-AGF	4.529 ± 0.013	7.011 ± 0.015
20 epochs, 1 seed	Temperature	w/o adaptive graph	2.105	3.193
		w/o gated fusion	2.716	3.665
		Full CLCRN-AGF	2.356	3.273
	Relative humidity	w/o adaptive graph	13.390	18.739
		w/o gated fusion	5.356	8.641
		Full CLCRN-AGF	5.151	8.097
	Surface wind	w/o adaptive graph	1.371	2.199
		w/o gated fusion	1.542	2.478
		Full CLCRN-AGF	1.483	2.319
	Cloud cover	w/o adaptive graph	0.1550	0.2511
		w/o gated fusion	0.1557	0.2551
		Full CLCRN-AGF	0.1559	0.2529

Values are mean±sample standard deviation for the three-seed blocks. The humidity w/o gated fusion run used automatic mixed precision for training; all reported test metrics use the same zero-inclusive evaluator.

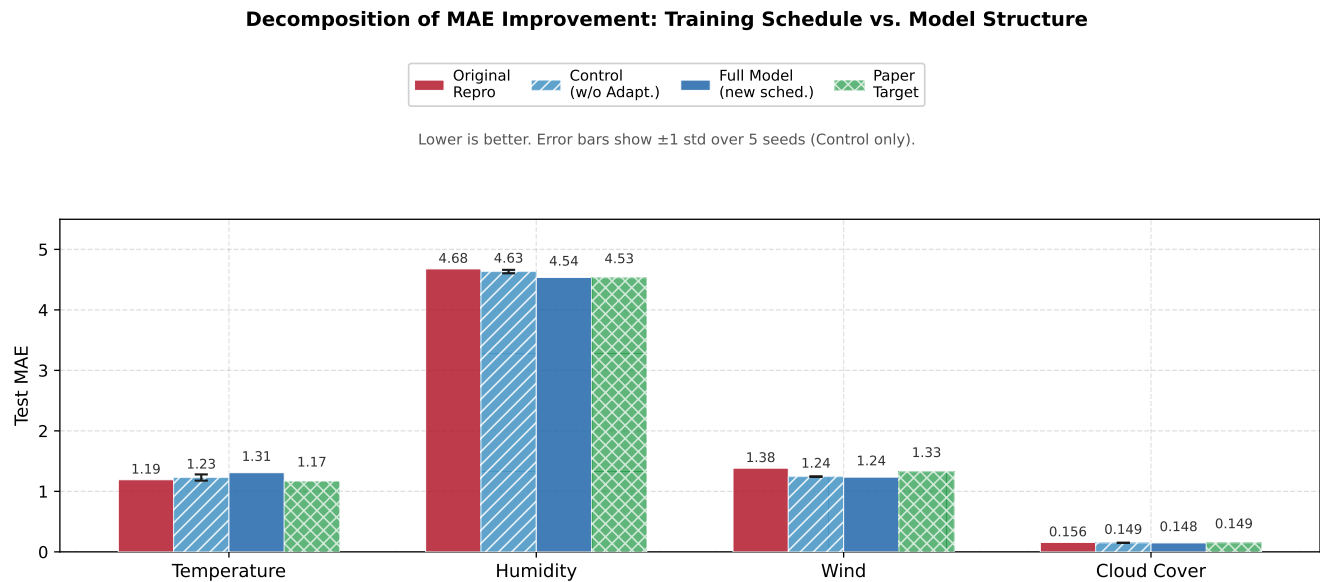


Figure 6. Decomposition of MAE into training-schedule and model-structure effects. Bars compare the original reproduction, the schedule-matched control, the full CLCRN-AGF model, and the published target. Error bars show ±1 standard deviation over five seeds for the control. The revised schedule explains most of the wind and cloud-cover gains; the adaptive branch adds a further reduction chiefly for humidity.

cover), with RMSE dispersion bounded by 0.047. Temperature is the most seed-sensitive variable—consistent with its smoother, more isotropic spatial field, for which optimization noise has a relatively larger influence on the final error—whereas humidity, wind, and cloud cover converge to nearly identical solutions once the schedule is fixed. The five-seed means also track the published CLCRN reference closely for humidity and cloud cover, indicating that the reported operating point is representative rather than a favourable single run.

Convergence behaviour reinforces the need for the longer schedule. Under the 100-epoch budget, the best validation checkpoint is reached on average at epoch 78 (temperature), 87 (humidity), 86 (wind), and 93 (cloud cover), and in several seeds approaches the budget ceiling. A short schedule would therefore systematically under-train the model and blur the separation between optimization and architectural effects that Fig. 6 is designed to expose; humidity additionally shows the widest seed-to-seed spread in stopping epoch, so convergence speed is itself task-dependent.

2.10 Robustness and computational cost

To probe robustness, Fig. 7 measures degradation under increasing missing-node ratios at inference, simulating sensor dropout on the spherical grid. Cloud cover is essentially insensitive, wind degrades slowly, and temperature and humidity degrade more steeply, mirroring the per-task error magnitudes in Fig. 5. On cost, the AGF encoder approximately doubles the trainable parameters (from 39.3k to 80.9k for temperature) and increases peak inference memory (from 352 to 635 MB) because it forms a dense $N \times N$ similarity matrix before top- k selection, while per-batch inference latency rises only marginally (202.6 to 211.9 ms). The extension is compact relative to global foundation models, but it is not cost-free and is not asymptotically linear in the number of nodes (Methods).

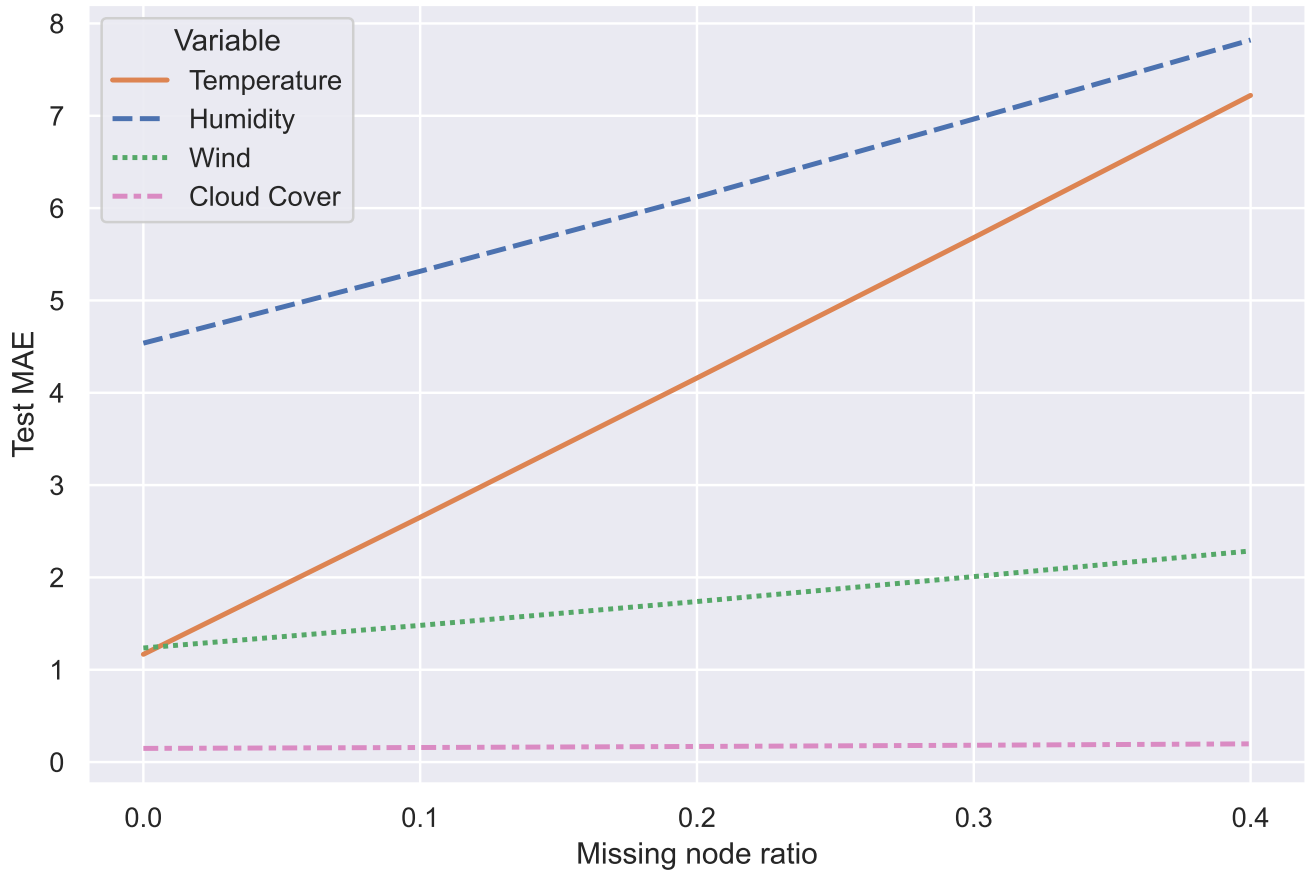


Figure 7. Robustness of the improved CLCRN under missing-node stress. Test MAE as a function of the fraction of nodes randomly masked at inference. Cloud cover is nearly flat; wind degrades slowly; temperature and humidity are more sensitive, consistent with their larger baseline error magnitudes.

2.11 Gate-activation diagnostics

The fusion gate $g \in (0, 1)$ controls, per node and channel, the balance between the graph-aggregated representation $\mathbf{q}$ and the residual/local representation $\mathbf{r}$ ($\mathbf{z} = g \odot \mathbf{q} + (1 - g) \odot \mathbf{r}$). Table 4 summarizes the learned gate values over the full test set, and Figure 8 shows their distribution (gate activations were exported for the three non-temperature variables). For each of these variables the mean gate is close to 0.5 (humidity $\mu = 0.55$, wind $\mu = 0.47$, cloud cover $\mu = 0.45$), so the model does not collapse onto either pathway; it uses both the non-local graph and the local residual. Humidity places slightly more weight on the graph branch and has the most concentrated distribution ($\sigma = 0.043$), consistent with its being the variable that benefits most from the adaptive graph; cloud cover has the widest spread ($\sigma = 0.106$), indicating a more sample-dependent balance. We report these statistics only as internal diagnostics: a non-degenerate gate distribution is consistent with, but does not by itself prove, a physical interpretation of either pathway.

Table 4. Gate-activation statistics over the test set (per-variable). The gate balances the graph-aggregated ($\mathbf{q}$) and residual ($\mathbf{r}$) representations; values near 0.5 indicate balanced use of both pathways.

Variable	Mean μ	Std σ	Min	Max
Relative humidity	0.550	0.043	0.392	0.692
Surface wind	0.474	0.047	0.333	0.648
Cloud cover	0.451	0.106	0.229	0.758

Distribution of learned gate activations

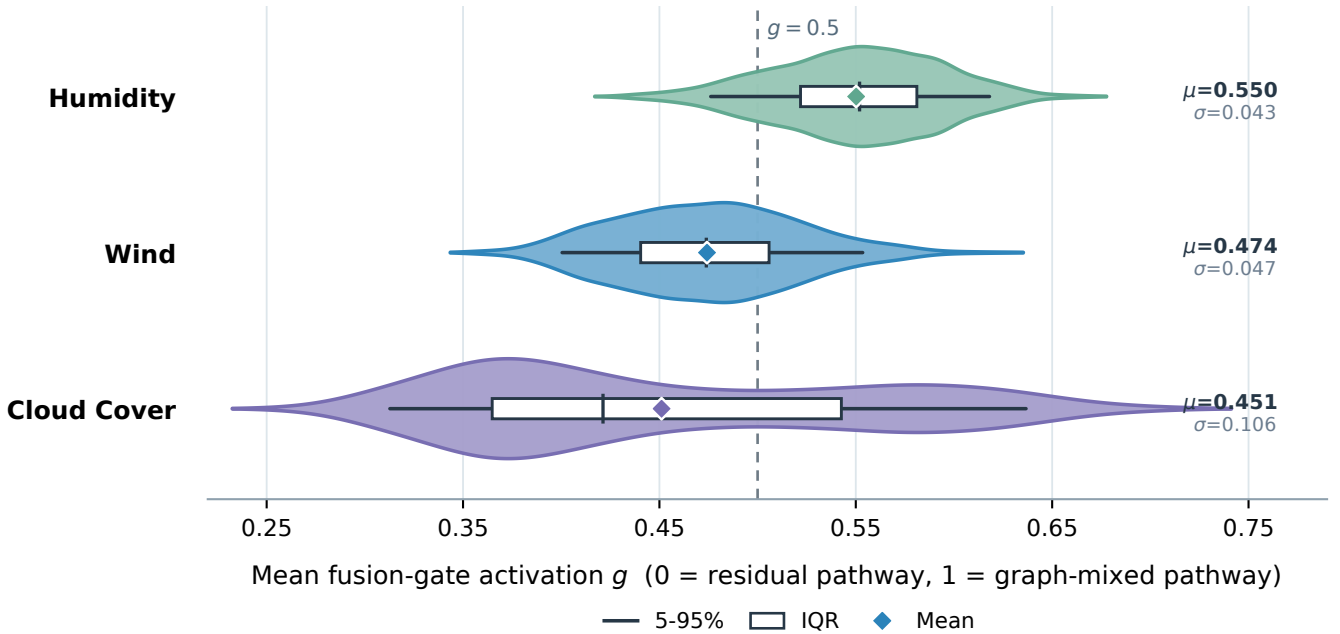


Figure 8. Distribution of learned fusion-gate activations. Violin plots of the gate value $g \in (0, 1)$ for three variables in a representative trained CLCRN-AGF model. The horizontal violin encodes density; the line spans the 5–95% interval, the white box marks the interquartile range, the internal tick marks the median, and the diamond marks the mean. The dashed reference line marks $g = 0.5$. All variables are centred near 0.5, indicating balanced use of the graph-aggregated and residual pathways; cloud cover’s wider spread reflects the intermittency of convective cloud formation. The plot is an internal diagnostic and is not, on its own, evidence that either branch corresponds to a specific physical process.

3 Discussion

What does the replicate experiment establish? The replicate experiment supports a narrow conclusion. Adding adaptive graph fusion to CLCRN is beneficial for relative humidity under the tested setup: the improvement is consistent across all five

seed labels and the Welch confidence interval excludes zero for both MAE and RMSE. For temperature, wind, and cloud cover, the evidence does not support a general claim of architectural superiority.

Why does relative humidity benefit the most? This task dependence is plausible but should not be over-interpreted. Relative humidity may benefit from dependencies that are not captured by the fixed local neighbourhood alone, which is consistent with the long-range and adaptive-graph motivations in the broader spatio-temporal forecasting literature^{19,20,26,47,48}. The current experiment does not, however, demonstrate that the learned edges correspond to teleconnections or moisture corridors. Establishing such a claim would require spatial edge analysis, comparisons with meteorological indices, and perturbation tests.

Separating optimization from architectural effects. The study also separates optimization effects from architectural effects. The revised schedule substantially improves wind and cloud-cover results even without AGF, while the additional AGF gain is small (Fig. 6). Reporting only the best single run would obscure this distinction. The five-replicate design is therefore central to the present conclusions. The ablations further show that the two AGF components do not contribute uniformly. For humidity, removing the adaptive graph branch gives the weakest full-budget result, whereas mean fusion without the learned gate remains close to the full model. For temperature, mean fusion can also match or outperform the learned gate. The evidence therefore supports the adaptive graph branch more strongly than it supports a universal benefit from gated fusion, although the humidity gate comparison should be interpreted cautiously because one ablation variant used mixed-precision training.

Limitations and future work. Several limitations remain. First, the study inherits the compact single-variable setup of the original CLCRN paper rather than evaluating full-state medium-range weather prediction as in recent global forecasters^{5,6,9,10}. Second, the current baseline set contains persistence, the published CLCRN reference, and a schedule-matched control; modern graph and non-graph baselines^{28,35,38,39} still need to be retrained under the identical data and optimization pipeline. Third, the top- k graph is obtained from a dense $N \times N$ similarity matrix, giving quadratic intermediate cost; sparse or linear-attention formulations are a natural remedy. Fourth, full-budget component ablations are now available for temperature and relative humidity, but wind and cloud cover still rely on the shorter 20-epoch component screening and should not be over-interpreted. The humidity mean-fusion completion used mixed-precision training, so a fully precision-matched ablation grid would be useful before making stronger claims about the gate.

4 Conclusion

This work presents a controlled evaluation of an adaptive graph-fusion extension to the published CLCRN weather-forecasting model, with the central aim of measuring—rather than asserting—when such an extension helps. Through a schedule-matched control, five-seed replication, and significance testing, the manuscript cleanly distinguishes prior CLCRN contributions, external published values, independent reproduction, and new experiments. Under this protocol, CLCRN-AGF consistently improves relative-humidity forecasting, while changes for temperature, wind, and cloud cover are small or metric-dependent, and a schedule-versus-structure decomposition shows that much of the apparent gain over the published reference is attributable to the optimized training schedule rather than the architecture. The results therefore support adaptive graph fusion as a task-dependent extension rather than a universally better replacement for the original CLCRN encoder. Beyond this specific model, the study’s main takeaway is methodological: schedule-matched, multi-seed, significance-tested evaluation is a low-cost discipline that separates genuine architectural gains from optimization and reporting effects, and is directly transferable to other spatio-temporal forecasting systems.

5 Methods

5.1 Data and preprocessing

We use the same four WeatherBench-derived tasks and public preprocessing pipeline as the original CLCRN study^{2,11}: 2-m temperature (K), relative humidity (percentage points), 10-m surface wind components (u and v , m s^{-1}), and total cloud cover (fraction). Hourly ERA5 records from 1 January 2010 through 31 December 2018 are mapped to a 32×64 latitude–longitude grid with $N = 2,048$ nodes. We denote the spherical grid by

$$\mathcal{V} = \{(\lambda_i, \phi_i)\}_{i=1}^N, \quad N = 32 \times 64 = 2,048, \quad (1)$$

where λ_i and ϕ_i are longitude and latitude.

Each example contains 12 consecutive hourly observations and targets the next 12 hours. Starting windows are sampled once per day (step size 24), producing 3,286 examples. The split is chronological and unshuffled: 2,300 training examples (70%), 329 validation examples (10%), and 657 test examples (20%). Input and target variables are standardized with means

and standard deviations computed from the training subset only; metrics are calculated after inverse transformation. For a starting index s , the input–target pair is

$$\mathbf{X}^{(s)} = [\mathbf{x}_s, \dots, \mathbf{x}_{s+T-1}], \quad \mathbf{Y}^{(s)} = [\mathbf{x}_{s+T}, \dots, \mathbf{x}_{s+T+H-1}], \quad T = H = 12, \quad (2)$$

with $\mathbf{x}_t \in \mathbb{R}^{N \times C}$. Standardization uses training statistics,

$$\bar{\mathbf{x}}_t = (\mathbf{x}_t - \boldsymbol{\mu}_{\text{train}}) \oslash \boldsymbol{\sigma}_{\text{train}}, \quad (3)$$

where $\oslash$ denotes element-wise division.

5.2 CLCRN backbone

The backbone is the public CLCRN implementation of Lin et al.¹¹. CLConv generates a location-dependent kernel from each centre–neighbour coordinate pair and multiplies it by geodesic-distance and orientation factors. The resulting operator replaces the linear transforms in a graph-convolutional GRU and is used in a sequence-to-sequence encoder–decoder over $T = 12$ input steps and $H = 12$ forecast steps with autoregressive teacher forcing. We retain this backbone without claiming its architecture as a contribution of the present study. For node i and neighbour j , the geometric descriptor can be written as

$$\boldsymbol{\xi}_{ij} = [\lambda_i, \phi_i, \lambda_j, \phi_j, d_{ij}, \theta_{ij}], \quad (4)$$

where d_{ij} is geodesic distance and θ_{ij} is relative orientation. The local CLConv kernel is therefore location dependent,

$$\mathbf{K}_{ij}^{(\ell)} = f_\ell(\boldsymbol{\xi}_{ij}) \rho_d(d_{ij}) \rho_\theta(\theta_{ij}), \quad (5)$$

where f_ℓ is the learned kernel generator for propagation view ℓ and ρ_d, ρ_θ denote the distance and orientation modulations. For node features $\mathbf{h}_j$, the CLConv aggregation is

$$\text{CLConv}(\mathbf{h})_i = \sum_{\ell} \sum_{j \in \mathcal{N}(i)} \mathbf{K}_{ij}^{(\ell)} \mathbf{h}_j, \quad (6)$$

with $\mathcal{N}(i)$ denoting the local spherical neighbourhood. In the recurrent backbone, CLConv replaces the affine maps in the GRU gates:

$$\begin{aligned} \mathbf{r}_t &= \sigma(\text{CLConv}_r([\mathbf{x}_t \parallel \mathbf{h}_{t-1}])) , \\ \mathbf{u}_t &= \sigma(\text{CLConv}_u([\mathbf{x}_t \parallel \mathbf{h}_{t-1}])) , \\ \tilde{\mathbf{h}}_t &= \tanh(\text{CLConv}_c([\mathbf{x}_t \parallel (\mathbf{r}_t \odot \mathbf{h}_{t-1})])) , \\ \mathbf{h}_t &= \mathbf{u}_t \odot \mathbf{h}_{t-1} + (1 - \mathbf{u}_t) \odot \tilde{\mathbf{h}}_t. \end{aligned} \quad (7)$$

5.3 Adaptive graph-fusion encoder

The AGF encoder is applied to the embedded node features before the recurrent backbone (Fig. 2a; the graph construction is detailed in Fig. 2c). The per-node feature is formed by concatenating a projected weather feature with a learned node embedding:

$$\mathbf{F}_{bti} = \phi_x(\bar{\mathbf{x}}_{bti}) \parallel \mathbf{e}_i^{\text{node}}, \quad \mathbf{F} \in \mathbb{R}^{B \times T \times N \times d}. \quad (8)$$

Trainable source and destination embeddings $\mathbf{E}_s, \mathbf{E}_t \in \mathbb{R}^{N \times d_e}$ ($d_e = 10$) define a directed adjacency

$$A_{ij} = \frac{\exp(\mathbf{e}_i^{s\top} \mathbf{e}_j^t)}{\sum_{\ell=1}^N \exp(\mathbf{e}_i^{s\top} \mathbf{e}_\ell^t)}, \quad (9)$$

equivalent to applying a row-wise softmax to $\mathbf{E}_s \mathbf{E}_t^\top$. For each node the implementation retains the self-edge and the $k - 1$ largest remaining entries, then renormalizes the selected weights:

$$\mathcal{T}_k(i) = \{i\} \cup \arg \text{top}_{j \neq i, k-1} A_{ij}, \quad \tilde{A}_{ij} = \frac{A_{ij}}{\sum_{\ell \in \mathcal{T}_k(i)} A_{i\ell}}, \quad j \in \mathcal{T}_k(i). \quad (10)$$

With $k = 4$, the graph-mixed feature is

$$\mathbf{m}_{bti} = \sum_{j \in \mathcal{T}_k(i)} \tilde{A}_{ij} \mathbf{F}_{btj}. \quad (11)$$

The graph-aggregated and residual features are projected separately,

$$\mathbf{q}_{bti} = \mathbf{W}_m \mathbf{m}_{bti} + \mathbf{b}_m, \quad \mathbf{r}_{bti} = \mathbf{W}_r \mathbf{F}_{bti} + \mathbf{b}_r, \quad (12)$$

and the learned gate is

$$\mathbf{g}_{bti} = \sigma(\mathbf{W}_g [\mathbf{r}_{bti} \parallel \mathbf{q}_{bti}] + \mathbf{b}_g). \quad (13)$$

The fused AGF output is

$$\mathbf{z}_{bti} = \mathbf{g}_{bti} \odot \mathbf{q}_{bti} + (1 - \mathbf{g}_{bti}) \odot \mathbf{r}_{bti}. \quad (14)$$

The fused output $\mathbf{z}$ (dimension 8) is concatenated with the standard CLCRN encoder inputs before the unchanged recurrent encoder. The schedule-matched control disables the entire AGF encoder, removing both the adaptive top- k mixing and its gate; the ablations in Table 3 replace the gate by the mean $\frac{1}{2}(\mathbf{q}_i + \mathbf{r}_i)$ (*w/o gated fusion*) or disable the adaptive branch (*w/o adaptive graph*).

5.4 Training configuration

Both variants use two recurrent layers, 32 hidden units, an 8-dimensional weather-feature embedding, an 8-dimensional CLConv embedding, one CLConv propagation view, and curriculum learning in the autoregressive decoder. CLCRN-AGF uses $d_e = 10$ -dimensional adaptive source/destination node embeddings, an 8-dimensional AGF output, and $k = 4$ selected neighbours including the self-edge. Training uses Adam with learning rate 0.01, batch size 32, gradient-norm clipping at 5, and a MultiStepLR schedule with milestones $\{10, 20, 30, 40, 50\}$ and multiplicative factor 0.95. Models train for at most 100 epochs and the checkpoint with the lowest validation MAE is used for test evaluation.

Experiments were run with Python 3.13, PyTorch 2.10, CUDA 12.8, and an NVIDIA GeForce RTX 5090 Laptop GPU. Seeds 2021–2025 are applied to NumPy and all PyTorch random-number generators. We do not claim bitwise deterministic reproducibility across hardware or library versions. Primary control and CLCRN-AGF runs were trained in standard precision. The additional relative-humidity w/o gated fusion ablation in Table 3 was completed with PyTorch automatic mixed precision and batch size 32 because standard-precision attempts repeatedly failed with CUDA allocator exhaustion on the available Windows GPU.

5.5 Metrics and statistical analysis

MAE and RMSE are computed after inverse scaling. The corrected evaluator includes all finite target values; zero is not treated as missing because zero wind components and zero cloud cover are physically meaningful. Persistence repeats the final input observation over the complete forecast horizon. The primary analysis compares five seed replicates for each variant. For each metric we report the mean and sample standard deviation across seeds, the mean difference, a two-sided 95% Welch confidence interval, and a two-sided Welch test. All reported test rows use this zero-inclusive evaluator. For archived runs with existing checkpoints, validation selection was recomputed with the same corrected MAE among the saved checkpoints before test evaluation. For M finite target entries and seed-level metric values m , the reported quantities are

$$\begin{aligned} \text{MAE} &= \frac{1}{M} \sum_{a=1}^M |\hat{y}_a - y_a|, & \text{RMSE} &= \left(\frac{1}{M} \sum_{a=1}^M (\hat{y}_a - y_a)^2 \right)^{1/2}, \\ \Delta_m &= \bar{m}_{\text{AGF}} - \bar{m}_{\text{ctrl}}, & t_W &= \frac{\bar{m}_{\text{AGF}} - \bar{m}_{\text{ctrl}}}{\sqrt{s_{\text{AGF}}^2/n_{\text{AGF}} + s_{\text{ctrl}}^2/n_{\text{ctrl}}}}. \end{aligned} \quad (15)$$

5.6 Computational complexity

The original CLConv implementation materializes dense $N \times N$ kernel operators, and the AGF encoder forms a dense $N \times N$ similarity matrix $\mathbf{E}_s \mathbf{E}_t^\top$ before top- k selection. The implementation therefore has quadratic intermediate memory and similarity-computation cost in N , despite the sparse aggregation that follows selection. Claims of linear scaling are not made.

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7 Author Contributions

X.G. conceived the extension, implemented the software, conducted the experiments and analysis, prepared the visualizations, and wrote the original draft. H.-Y.L. supervised the study and reviewed and edited the manuscript. Both authors reviewed and approved the final manuscript.

8 Data Availability

The raw ERA5 variables are available through the [WeatherBench GitHub repository](#) and the Copernicus Climate Data Store. The original CLCRN project also provides preprocessed WeatherBench files through the data link described in its public README. The preprocessing period, variables, temporal windows, and chronological split used here are specified fully in the Methods, so the exact arrays can be regenerated deterministically from the public ERA5/WeatherBench sources. Because the preprocessed arrays are derived from third-party data whose licence governs redistribution, we provide the preprocessing and split scripts together with fixed configuration files in the public code repository and the deposited review archive, rather than redistributing the third-party arrays themselves. The archive additionally contains the corrected metric tables and per-seed summaries needed to reproduce every reported table and figure.

9 Code Availability

The study uses the public CLCRN implementation, extended with the AGF module and the corrected evaluation scripts described in this manuscript. The source files, run configurations, corrected evaluation script, humidity component-ablation completion script, and per-seed result summaries are included in the supplementary software/review archive submitted with this manuscript. The same materials are deposited in a public GitHub repository ([gengx-prog/CLCRN-AGF-Scientific-Reports](#)), with the review archive attached to release [v0.1-review](#).

10 Competing Interests

The authors declare no competing interests.

11 Generative AI and AI-Assisted Technologies Disclosure

During preparation of this manuscript, the authors used AI-assisted language editing to improve readability and organization. The authors reviewed and edited all generated content and take full responsibility for the final manuscript.

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